Model and Data Poisoning

4.2.3

SECAI+ PREP

AI powers critical systems from healthcare to finance but its reliability depends on the integrity of its training data and code. This lesson explores how model poisoning works, when it happens, and why it matters.

What Is Model Poisoning?

Any attack that **corrupts an AI model** is considered model poisoning. It most commonly occurs by manipulating the training dataset but it can strike at any stage of the AI lifecycle.

Malicious Data Injection

Attackers insert mislabeled or harmful data into the training set to alter what the model learns.

Incorrect Associations

The model learns distorted patterns and makes dangerous or incorrect decisions in production.

Hard to Detect

Poisoned models may appear accurate in most cases while failing catastrophically on specific inputs.

Poisoning Before or During Training

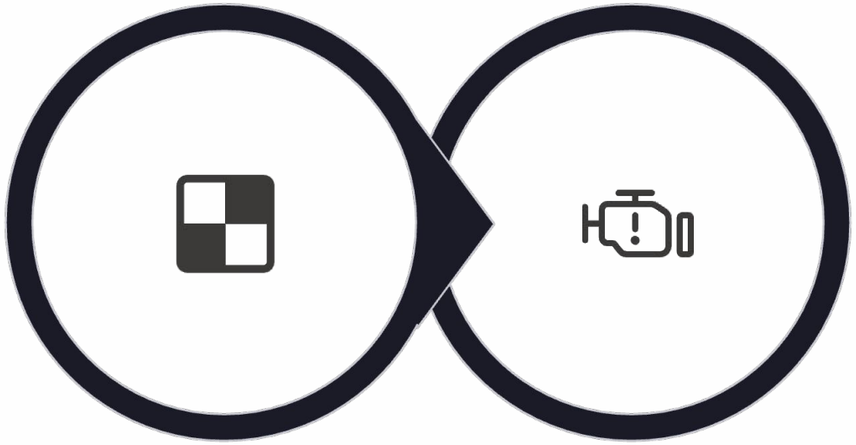
How It Works

Attackers inject **malicious or mislabeled data** into the training set before or during the training process. The model learns from this distorted data and behaves incorrectly when deployed in the real world.

Real-World Example

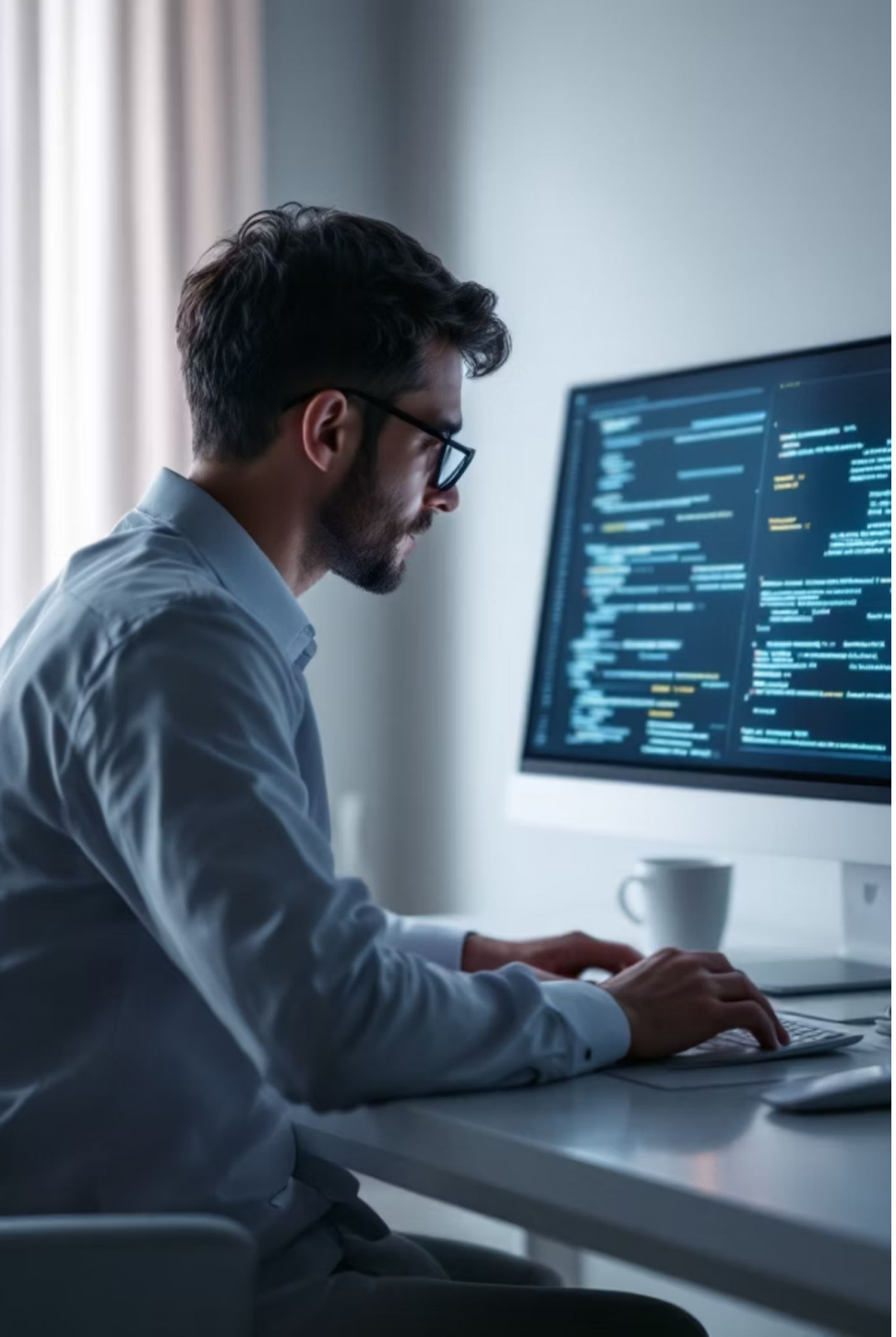
In a self-driving car AI, photos of dogs are mislabeled as "**speed limit 70 mph.**"

Result: the car **accelerates** when it sees a dog instead of slowing down — a potentially fatal outcome caused entirely by corrupted training data.



Inject Data

Model Fails



Poisoning After Training

Models can also be poisoned **after training is complete** even if the original training data was clean and trustworthy.

1. **Clean Training**
   The model is trained on good, uncontaminated data and performs correctly.
2. **Source Code Modified**
   An attacker directly modifies the model's source code or saved artifacts after training.
3. **Bad Behavior in Production**
   The model now follows malicious instructions regardless of what it originally learned.

This highlights the need to protect **model artifacts post-training**, not just the data pipeline. Security must extend beyond the training phase.

Why This Matters

Poisoned models behave unexpectedly in ways that are **hard to detect**. The model may appear accurate in most cases while failing dangerously on specific, targeted inputs.

Healthcare

Misdiagnosis or incorrect treatment recommendations from a poisoned medical AI.

Finance

Fraudulent transactions approved or legitimate ones blocked by a corrupted fraud detection model.

Transportation

Self-driving vehicles making dangerous decisions based on poisoned perception models.

Cybersecurity

Threat detection systems blinded to specific attack patterns by adversarial data poisoning.

Key Takeaways

1 **Model Poisoning Defined**

Any attack that corrupts an AI model's behavior — regardless of when or how it occurs.

2 **Training Data Is the #1 Attack Vector**

Mislabeled or malicious data injected during training shapes what the model learns — with dangerous consequences.

3 **Post-Training Attacks Are Equally Dangerous**

Direct source code modification after training is often overlooked but just as harmful.

4 **Protect the Entire AI Lifecycle**

Both training pipelines *and* model artifacts must be secured — poisoning can happen at any stage.